

ABDULLAH AL MAZID

Industrial & Production Engineering Graduate

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Career Objective

To pursue a lecturer position in Industrial & Production Engineering where I can contribute to undergraduate teaching, student mentoring, academic research, and laboratory-based learning. I aim to apply my background in operations research, quality engineering, manufacturing systems, data analytics, and intelligent manufacturing to support effective classroom instruction, practical engineering education, and research-oriented academic development.

Education

Bangladesh University of Engineering & Technology (BUET)

2022–2026

CGPA: 3.67/4.00

B.Sc. in Industrial & Production Engineering

Chattogram Collegiate School and College

2018–2020

GPA: 5.00/5.00

Higher Secondary Certificate (HSC)

Teaching Interests

Quality Control Management | Operations Research | Supply Chain Management | Operations Management | Manufacturing Processes | Engineering Economics | Probability & Statistics | Computer Integrated Manufacturing | Intelligent Manufacturing | Product Design

Research Publications

- [1] **eGov-Lens: Multi-Dimensional ML for Aspect-Based Bengali Public Feedback Analysis**, *ICCIT 2025, IEEE*. DOI: 10.1109/ICCIT68739.2025.11491291
- [2] **Stochastic Modeling of Throughput-Quality Dynamics in Labor-Intensive Manufacturing**, *IEOM 2025*. DOI: 10.46254/BA08.20250379
- [3] **Validation of Temperature Effect on 3D Printed PLA Materials using FEA & Hyperelastic Modeling**, *IEOM 2024*. DOI: 10.46254/BA07.20240046

Selected Academic Projects

AutoPrompt-SAM — Undergraduate Thesis

Computer Vision / Defect Detection

- Developed a YOLOv11-guided automatic segmentation framework for defect detection in CFRP thermography, achieving IoU and Dice scores of 1.0000 on held-out test data.
- Applied machine learning and computer vision concepts to automated quality inspection in advanced manufacturing materials.

Operations Research Optimization — Cafeteria Production Planning

Optimization / Production Planning

- Developed a linear programming model to optimize production quantity and pricing under demand, resource, and operational constraints using Gurobi and Excel Solver.

IoT Smart Conveyor System for Quality Inspection

Automation / Quality Control

- Built an automated defect detection and inventory tracking system using ESP32, sensors, actuators, and a real-time monitoring dashboard.
- Integrated process control and alerting features for production-line quality assurance and monitoring.

- Designed a commercial and healthcare-oriented cleaning system; performed ANSYS stress and fatigue analysis with break-even and sensitivity analysis.

Technical and Teaching Skills

Core IPE Areas: Operations Research, Production Planning, Inventory Control, Statistical Quality Control, Manufacturing Systems.

Teaching Support: Lecture preparation, academic presentation, technical writing, student mentoring, laboratory demonstration.

Engineering Software: SolidWorks, ANSYS Mechanical, AutoCAD, CATIA, Arena Simulation, MATLAB.

Programming & Analytics: Python, NumPy, pandas, scikit-learn, PyTorch, Excel, Power BI, SQL.

Machine Learning & IoT: Computer Vision, YOLOv11, SAM, NLP, ESP32/Arduino, real-time monitoring dashboards.

Industrial Exposure

Industrial Attachment — PRAN-RFL Group 05 Nov–05 Dec 2025 / Dhaka, Bangladesh

- Observed large-scale FMCG production across beverage, dairy, biscuit, and confectionery lines, gaining practical exposure to production flow, quality practices, and supply chain operations.
- Studied time-and-motion analysis, production scheduling, statistical quality control, industrial automation, operational control, and inventory management practices in a live manufacturing environment.

Achievements

- **Deen Scholarship Recipient** — Two-time merit-based award recognizing academic excellence at BUET.
- Three peer-reviewed conference publications across IEEE and IEOM as an undergraduate student.
- Completed applied academic projects involving quality inspection, production planning, automation, optimization, and process improvement.

Language Proficiency

Bengali: Native **English:** Professional working proficiency in reading, writing, speaking, and academic communication

References

Dr. Abdullahil Azeem
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Relationship: Thesis Supervisor

Dr. Syed Mithun Ali
Professor, Department of Industrial & Production Engineering, BUET
Email: mithun@ipe.buet.ac.bd
Relationship: Course Instructor